Harsh Shah

LinkedIn | Email | GitHub |+1-716-939-8261

Senior Software Engineer with 8 years of experience designing high-throughput microservices and distributed systems capable of processing 14M+ daily transactions. Expert in modernizing complex architectures through AI-assisted development

Technologies

- **Languages & Core:** Java (8/11/17), Python, SQL
- **Distributed Systems:** Apache Kafka, Spring Boot, Microservices, REST, Protobufs
- **Data Stores:** MongoDB, Hazelcast, Distributed RDBMS (Sybase/PostgreSQL)
- **AI & Productivity:** GitHub Copilot, OpenCode, Prompt Engineering

Education – Masters in computer science from State University of New York at Buffalo

Certification – AWS Certified Cloud Practitioner (View Credential)

Senior Software Engineer at Goldman Sachs, NYC | August 2021 to present | 5 years

Project: Event-Driven Data Platform
Technologies: Java 11, React, Apache Kafka, MongoDB, Hazelcast (Centralized cache)

- Architected the migration of high-volume data streams from a monolithic legacy system to a modern, event-based microservices architecture, reducing end-to-end processing latency by 2 hours during peak loads
- Optimized message distribution patterns by implementing a multi-partition Kafka strategy and custom partition keys, successfully eliminating consumer lag and achieving a peak throughput of **200K+ messages/hour**
- Built robust system resilience by implementing **distributed back off and retry patterns** for external API integrations, significantly reducing processing failures.
- **Automated Regression:** Developed a high-fidelity regression testing suite that captures production Kafka traffic and replays it in QA.
- **Integrated AI-based log analysis** tools to ingest and compare logs from previous releases against current runs, utilizing LLMs to detect anomalous behavioral shifts and ensure no fundamental breaking changes in complex business flows.

Project: Securities Trade Confirms & Digital Delivery Platform
Technologies: Java 17, Spring Boot, Kubernetes (K8s), AWS, Apache Kafka, REST APIs

- **Designed and implemented** a high-availability microservices architecture using Spring Boot and Java 17, exposing secure RESTful APIs to facilitate seamless cross-platform data exchange.
- Standardized API documentation and contract-first development using Swagger/OpenAPI, improving developer experience and reducing integration time for downstream consumer teams
- Utilized JUnit and Mockito for robust unit testing and WireMock for mocking HTTP-based services, ensuring high code coverage and system reliability during the migration.

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Project: Distributed Stream Processing & Configuration Service
Technologies: Java 11, Apache Kafka, Socket Programming, Python, MongoDB

- Developed a **low-latency data bridge** using Socket Programming to interface between legacy data stores and Apache Kafka, enabling real-time stream processing and reducing data ingestion lag
- Collaborated to migrate core processing logic from IBM Mainframe systems to Sequencer Based Architecture, resulting in a modern platform capable of reliably **handling 13-14 million** stock exchange transactions daily across global markets
- Engineered a **scalable configuration management service**, replacing a legacy internal tool with a modern UI/API framework that automated onboarding workflows and implemented distributed audit logging for system-wide transparency

Project: Distributed Transaction & Workflow Orchestration
Technologies: Java 8, Spring Boot, Apache Kafka, REST

- Architected an end-to-end **workflow orchestration engine** that eliminated manual intervention in Payment Processing Platform, reducing operational risk and reclaiming **20+ engineering/ops hours per week**
- **Engineered dynamic schema updates** within the transaction pipeline to support evolving regulatory standards, ensuring 100% data integrity and backward compatibility across high-volume data formats

Software Engineer at Amdocs, India | September 2016 to January 2020 | 3 ½ year

Project: Order Management System – Digital Transformation
Technologies: Java, REST API, SQL Query Tuning

- Acted as SME for the design and implementation of production-grade RESTful web services in Java, enabling reliable data consumption for mobile and web platforms.